



Curriculum Council
Government of Western Australia



WACE Examination, Sample 2008

ENGINEERING STUDIES

Section 2, Electrical, electronic systems

Stage 3

Provisional Marking Guidelines

Section 2: Electrical, electronic systems

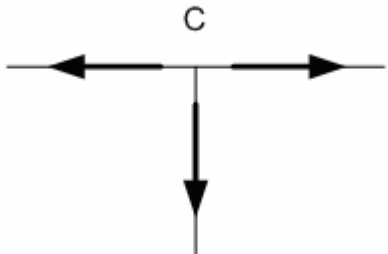
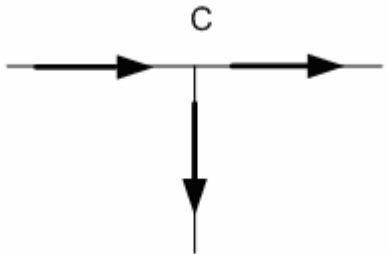
Part A: Multiple Choice Questions

1	A potential difference across a power supply or component	1
2	B 5V	1
3	D adjust the voltage across the components	1
4	A create an open circuit if the current is too large	1
5	C LDR	1
6	D see (D) diagram on exam paper	1
7	B protects components from current flowing in the wrong direction	1
8	C 12V	1
9	D 42 μF	1
10	A 3.43 Ω	1
		Total 10

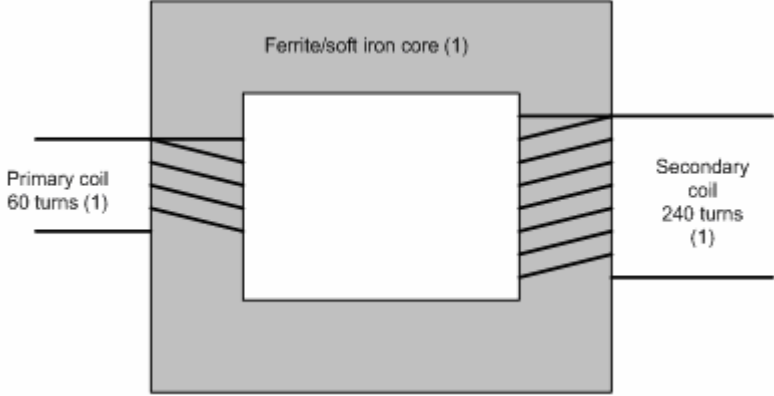
Part B: Written

Section 2 — Electrical, electronic systems		
1a	(i) $R_1 = 20 + 20$ $= 40\Omega$ $R_2 = R_1 \parallel R$ Therefore: $\frac{1}{R_2} = \frac{1}{R_1} + \frac{1}{R}$ $= \frac{1}{40} + \frac{1}{120}$ $= \frac{4}{120} = \frac{1}{30}$ Therefore: $R_2 = 30\Omega$ Finally: $R_T = 10 + R_2$ $= 10 + 30 = 40\Omega$	1 1 1 1 1 1 1 1 TOTAL 8
	(ii) $12 = R_T I_1$ $\therefore I_1 = \frac{12}{R_T} = \frac{12}{40} = 0.3 A$	1 1 TOTAL 2
	(iii) $P_s = 12 \times I_1$ $\therefore P_s = 12 \times 0.3 = 3.6 W$	1 1 TOTAL 2

	(iv) There are numerous ways to solve for V_C. A simple way is to make use of the resistance calculations and the resistor-divider rule. The equivalent resistance across node C and ground is $R_2 = 30\Omega$. Now using the resistor divider rule: $V_C = \frac{R_2}{R_2 + 10} \times 12$ $= \frac{30}{30 + 10} \times 12$ $= 9V$	2 1 1 TOTAL 4
	(v) $V_D = \frac{20}{20 + 20} \times V_C$ $= \frac{20}{40} \times 9$ $= 4.5V$ Note: Use the answer from (iv) for V_C	1 1 TOTAL 2
1b	$V_D = 20 \times I$ $\therefore I = \frac{V_D}{20}$ $\therefore I = \frac{2}{20} = 0.1A$ Voltage at node C: $V_C = 20 \times I + 20 \times I$ $= 20 \times 0.1 + 20 \times 0.1 = 4V$ Current through 10Ω resistance: $12 = 10 \times I_1 + V_C$ $\therefore I_1 = \frac{12 - V_C}{10} = \frac{12 - 4}{10} = 0.8A$ Current through R: $I_1 = I_R + I$ $\therefore I_R = I_1 - I = 0.8 - 0.1 = 0.7A$ Resistance R: $V_C = R \times I_R$ $\therefore R = \frac{V_C}{I_R} = \frac{4}{0.7} = 5.714\Omega$	1 1 1 2 1 1 1 1 TOTAL 10
1c	The sum of the currents leaving node C equals zero: the current flowing to the left through the 10Ω resistor, plus the current flowing to the right through the 20Ω resistor, plus the current flowing down through R, equals zero.	1

	 Alternatively, the current that enters from the left branch is equal to the sum of the currents that leave the right and lower branches: 	1 TOTAL 2
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Section 2 — Electrical, electronic systems		
2a	(i) $Efficiency = \frac{Power\ out}{Power\ in} \times 100$ $\therefore 80 = \frac{100kW}{P_{in}} \times 100$ $\therefore P_{in} = \frac{100kW \times 100}{80} = 125\ kW$ Energy (in joules) = Power (watts) x time (seconds) Therefore: $E = P_{in} \times t$ $= 125000 \times (15 \times 60)$ $= 125000 \times 900$ $= 112.5\ MJ$	1 1 1 1 1 1 1 1 TOTAL 7
	(ii) Energy (in kWh) = Power (kW) x time (hours) Therefore: $E = P_{in} \times t$ $= 125 \times 7$ $= 875\ kWh$	1 1 1 TOTAL 3

2b	(i) Step-up	1 TOTAL 1
	(ii) <i>Turns ratio = voltage ratio</i> $\therefore \frac{N_2}{60} = \frac{V_2}{V_1} = \frac{12}{3}$ $\therefore N_2 = \frac{12}{3} \times 60 = 240 \text{ turns}$	1 1 TOTAL 2
	(iii) 	1 for primary coil 1 for secondary coil 1 for iron core TOTAL 3
	(iv) AC only. It's a time-changing primary current that induces a voltage across the secondary coil terminals. (DC is not a changing current.)	1 1 TOTAL 2
	(v) $V_2 = \frac{480}{60} \times 3$ $= 24 \text{ V}$ $I_2 = \frac{V_2}{10}$ $= \frac{24}{10}$ $= 2.4 \text{ A}$ For an ideal transformer: power out = power in, therefore $V_1 \times I_1 = V_2 \times I_2$ $3 \times I_1 = 24 \times 2.4$ $\therefore I_1 = 19.2 \text{ A}$	1 1 1 1 1 TOTAL 5
2c	(i) A signal from an electronic/electrical circuit (usually a low voltage circuit) causes a current to pass through the coil.	1

	This current induces a magnetic field about the soft iron core, pulling the armature down. This moves the common from the NC contact to the NO contact. This will either switch on or off an external circuit. When the coil is deactivated, the spring on the armature will pull the common back to the normally closed contract.	1 1 1 TOTAL 4
2c	(ii) Control circuits tend to use low voltage DC, and so cannot directly drive output transducers required by the application. An example is a 9VDC battery operated water reticulation computer controlling 24VAC solenoid valves.	1 1 1 TOTAL 3

Section 2 — Electrical, electronic systems		
3a	(i) From Kirchhoff's voltage law: $12 = 5000 \times I_R + 0.7$ $I_R = \frac{12 - 0.7}{5000} = 2.26 \times 10^{-3} A$ Now $I_{LDR} = 0$ Now, from Kirchhoff's current law: $I_B = I_R - I_{LDR}$ $= 2.26 \times 10^{-3} - 0$ $= 2.26 \times 10^{-3} A$ Finally, as the transistor is saturated, then $V_C = 0$. Now Kirchhoff's voltage law gives: $12 = 2 + R_D \times I_D + 0$ $\therefore I_D = \frac{12 - 2}{R_D} = \frac{10}{200} = 0.05 A \quad (50 mA)$	1 1 1 1 1 1 1 1 TOTAL 8
	(ii) $\frac{I_C}{I_B} = \frac{0.05}{2.26 \times 10^{-3}} = 22.1$ As $\frac{I_C}{I_B} < \beta$ then this confirms that the transistor is saturated.	1 1 TOTAL 2

3b	(i) As the LDR acts like a short circuit, the voltage drop across base-emitter is zero. Clearly this is less than 0.7V and so the transistor is off.	1 1 TOTAL 2
	(ii) The transistor is off so that $I_B = 0$ Kirchhoff's law: $12 = 5000 \times I_R + 0$ $\therefore I_R = \frac{12}{5000} = 2.4 \times 10^{-3} A \quad (2.4 mA)$ Now $I_{LDR} = I_R$ $\therefore I_{LDR} = 2.4 \times 10^{-3} A$	1 1 1 TOTAL 4
3c	(i) $I_{LDR} = \frac{V_{BE}}{350}$ $= \frac{0.7}{350} = 2.0 \times 10^{-3} A \quad (2.0 mA)$ Kirchhoff's voltage law $12 = 5000 \times I_R + V_{BE}$ $= 5000 \times I_R + 0.7$ $\therefore I_R = \frac{12 - 0.7}{5000} = 2.26 \times 10^{-3}$ Kirchhoff's current law $I_B = I_R - I_{LDR}$ $= 2.26 \times 10^{-3} - 2.0 \times 10^{-3} = 0.26 \times 10^{-3} A \quad (0.26 mA)$ With the transistor in its forward active region: $I_D = \beta \times I_B$ $= 100 \times 0.26 \times 10^{-3}$ $= 26 \times 10^{-3} A \quad (26 mA)$	1 1 1 1 1 1 TOTAL 8
3c	(ii) $12 = 2.0 + 200 \times 0.026 + V_C$ $\therefore V_C = 12 - 2.0 - 200 \times 0.026 = 4.8V$ As $V_C > V_{CE,sat}$ then the transistor is not saturated, and so is	1 1 1

	operating in its forward-active region.	TOTAL 3
3d	The maximum current that flows through the diode will occur when the transistor is saturated. So R_D is chosen to keep I_D equal or less than 20mA when the transistor is saturated (and so $V_C = 0$). Then $12 = 2.0 + R_D I_D + 0$ $\therefore I_D = \frac{12 - 2}{R_D}$ Now $\frac{12 - 2}{R_D} \leq 0.020$ $\therefore R_D \geq \frac{10}{0.02} = 500 \Omega$ So the smallest R_D is 500Ω.	1 1 1 TOTAL 3